



Communications and Networks

Lecture 5: Baseband Transmission

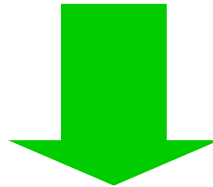
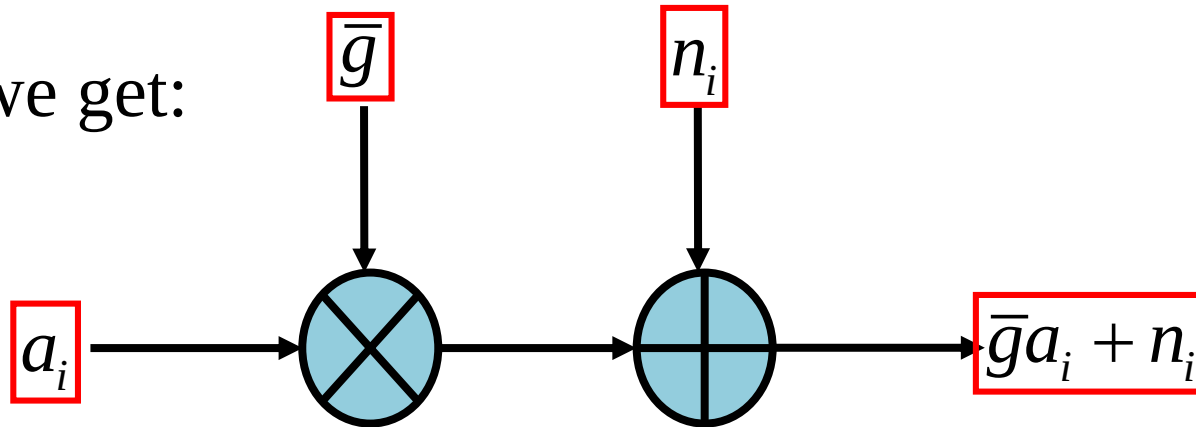
- Detection and Error Analysis

Sheng Zhou



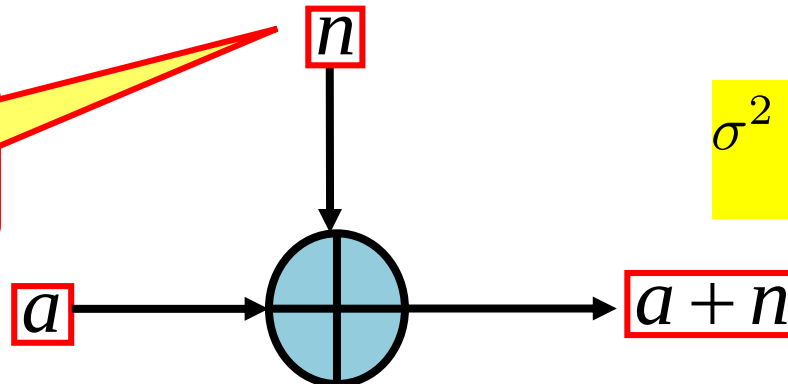
Equivalent Baseband Transmission Model

■ Finally we get:



$$n \sim \mathcal{N}(0, \sigma^2)$$

Coefficients are reflected in the noise power

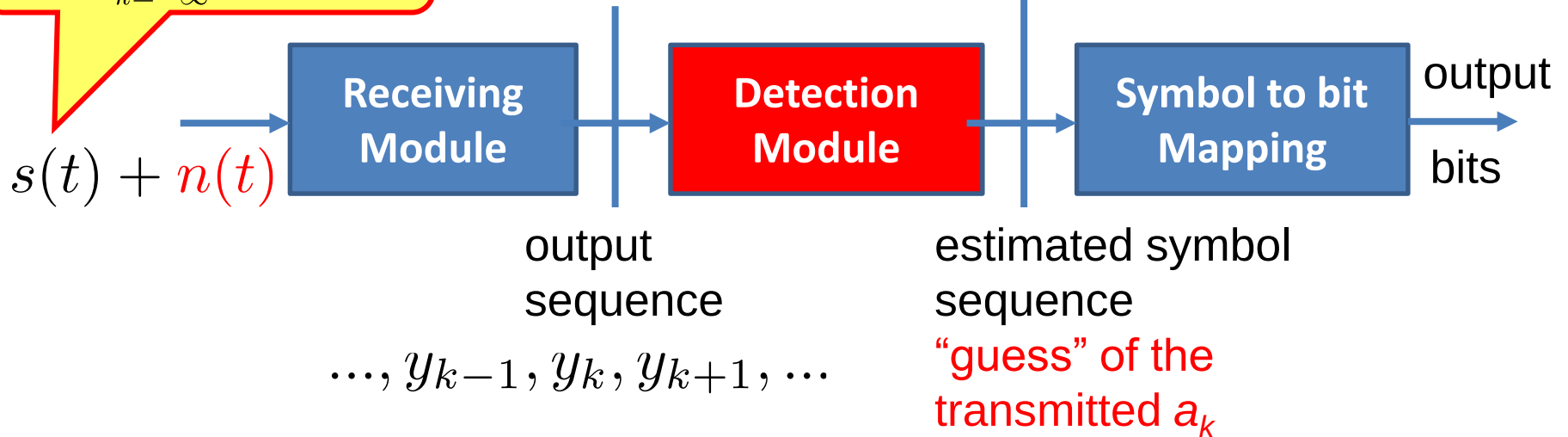


$$\sigma^2 = \frac{n_0}{2 \int_{-\infty}^{\infty} |G(f)|^2 df}$$



Baseband Demodulator: Components

$$s(t) = \sum_{k=-\infty}^{\infty} a_k g(t - kT_s)$$



Goal: Minimize the probability of erroneous guess

- Property of the noise (last lecture)
- How to design the receiving module (last lecture)
- How to make the symbol decision (this lecture)



Outline

- Optimal Detection
- Error Probability Analysis
- Revisit the Baseband Receiver



MAP Detection

- Maximize the a posteriori probability

$$P(a|y) = \frac{f(y|a)P(a)}{f(y)}$$

- We can drop $f(y)$, leading to:

$$f(y|a=A)P(a=A) \begin{matrix} > \\ < \end{matrix}^{\begin{matrix} A \\ -A \end{matrix}} f(y|a=-A)P(a=-A)$$

$$\frac{f(y|a=A)}{f(y|a=-A)} \begin{matrix} > \\ < \end{matrix}^{\begin{matrix} A \\ -A \end{matrix}} \frac{P(a=-A)}{P(a=A)}$$

Only need to
compare the
conditional
probability here



Maximum Likelihood (ML) Detection

- If we assume **equal probability of all symbols** (this assumption generally holds throughout this course)

- ▶ For the binary case $P(a = -A) = P(a = A)$

- ▶ Then we can transform the MAP rule to

$$\frac{f(y | a = A)}{f(y | a = -A)} \underset{-A}{\overset{A}{>}} 1$$

y looks like to whom among -A and A most?

- ▶ This is called the **Maximum Likelihood (ML)** rule
- ▶ Only when the probability of the hypothesis are the same, MAP is equivalent to ML



ML for the Additive Gaussian Channel

- The conditional probability is

$$f(y | a = A) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y - A)^2}{2\sigma^2}\right)$$

$$f(y | a = -A) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y + A)^2}{2\sigma^2}\right)$$

- With **additive Gaussian Noise**, we have the *minimum distance* detection rule

$$\frac{f(y | a = A)}{f(y | a = -A)} \underset{-A}{\overset{A}{>}} 1 \quad \Rightarrow \quad |y - A| \underset{A}{\overset{-A}{>}} |y + A| \quad \Rightarrow \quad \begin{matrix} A \\ y > 0 \\ -A \end{matrix}$$

Exactly as our previous intuition!



More about the Q function

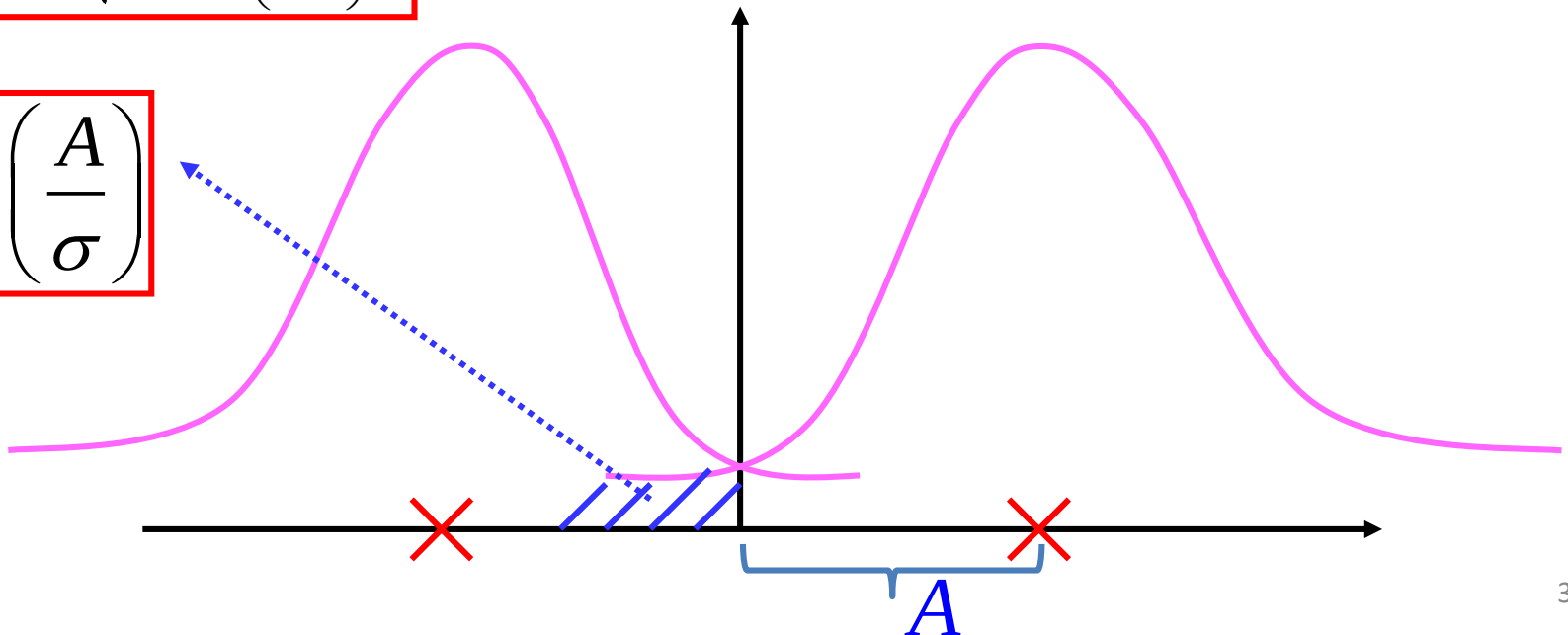
■ Generalization:

- For any symbol and its detection threshold with distance A from it, the probability that the received signal y is outside the threshold (leads to detection error) is

$$Q(x) = \int_x^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right) dt$$

$$Q\left(\frac{A}{\sigma}\right)$$

$$Q\left(\frac{A}{\sigma}\right)$$





Summarize the Steps of Calculating SEP

- 1. Draw the symbols
 - ▶ With inter-symbol distance $2A$
- 2. Decide the detection thresholds
 - ▶ Distance between the symbol to its thresholds is A
- 3. Derive the conditional error probability for each symbol **as the function of $\frac{A}{\sigma}$**
 - ▶ **Case by Case**
- 4. Use A to characterize S
 - ▶ Obtain A/σ as the function of S/N
- 5. Finally get the SEP as a function of S/N , which is $2E_s/n_0$

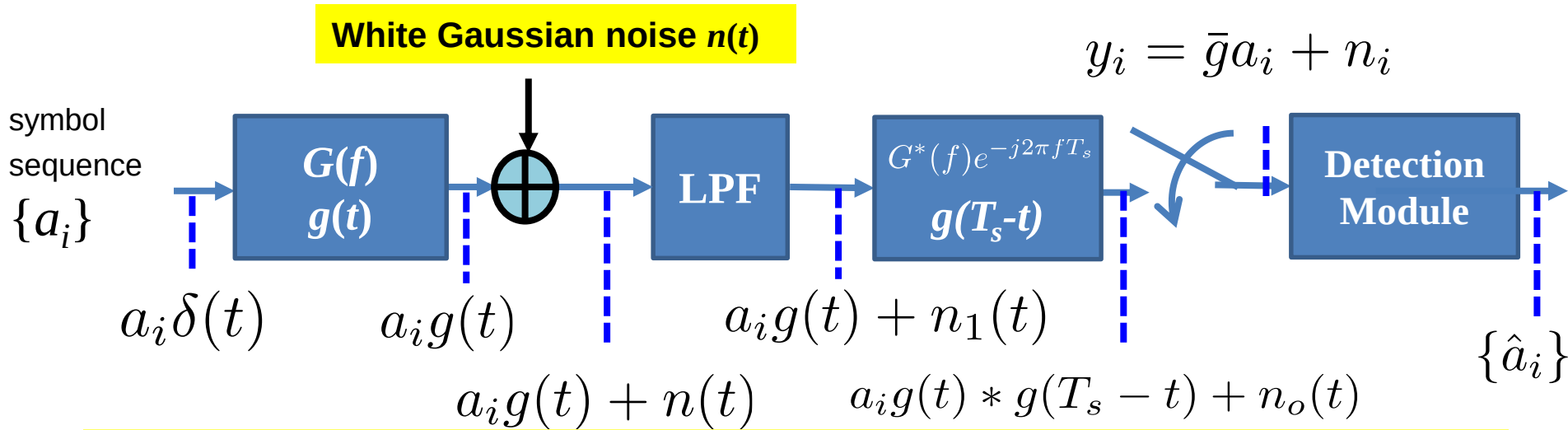


Outline

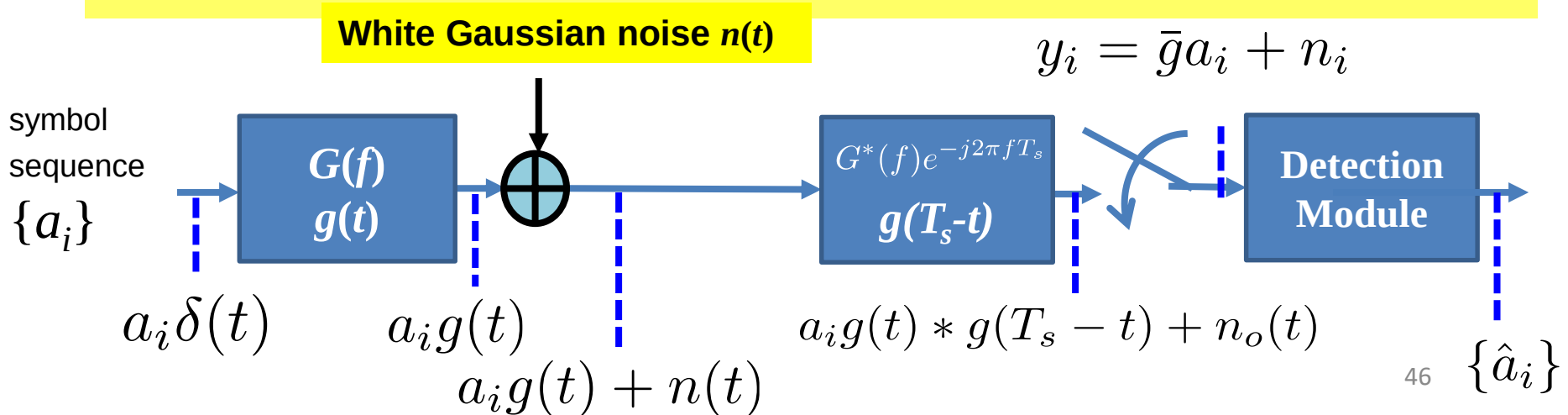
- Optimal Detection
- Error Probability Analysis
- Revisit the Baseband Receiver



Revisit the Baseband Receiver



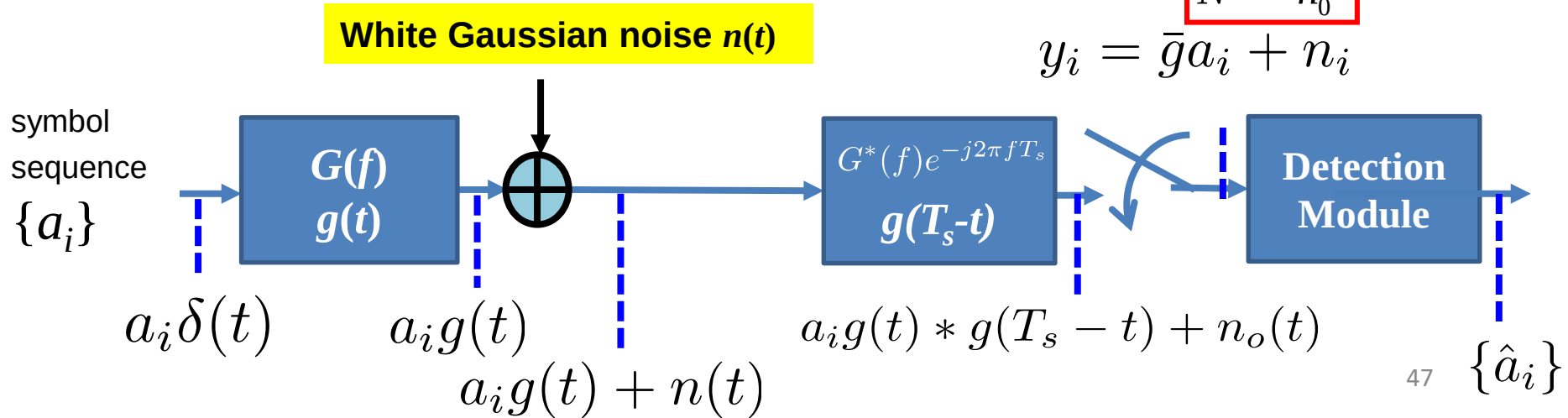
Can we simplify or integrate the blocks at the receiver side?





Revisit the Baseband Receiver

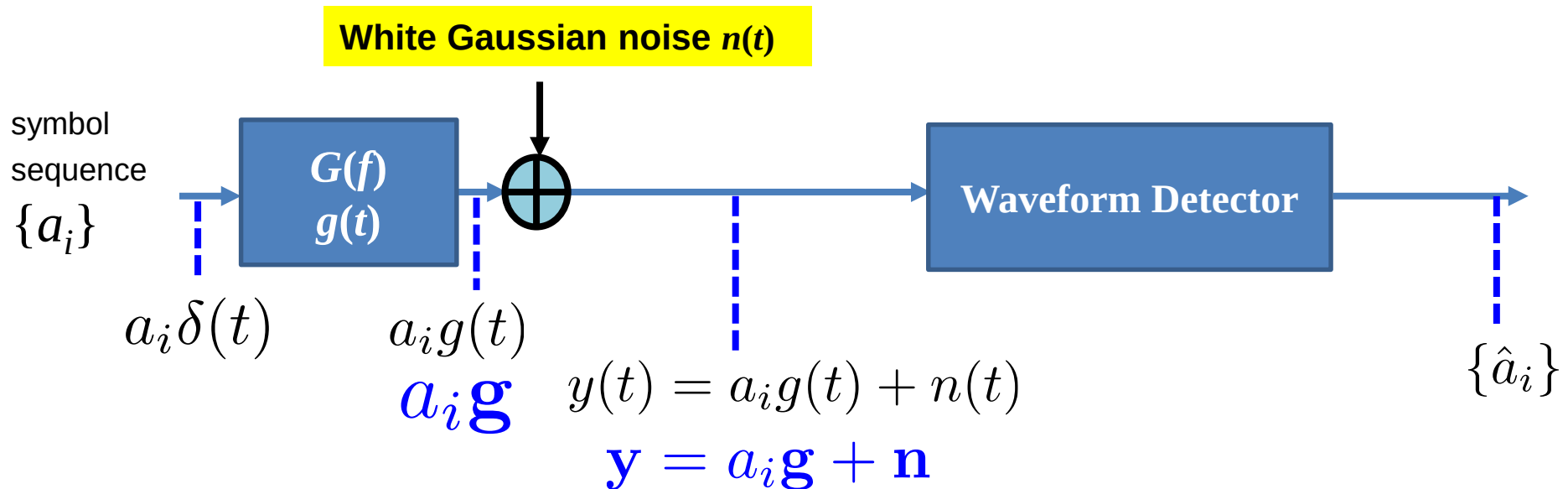
$$\frac{S}{N} = \frac{2E_s}{n_0}$$



- What we really care is the SEP
- But why we optimize the SNR by using M.F. in the first place?
- Is it **optimal to use the two separated M.F. and MAP detection blocks**? Or is it just some engineer simplification?



End-to-End “Waveform Detector”



- Consider an OPTIMAL “waveform detector” that directly guesses the symbol based on the received waveform $y(t)$
- **Approximate** the waveform by a vector (*spam over time with k elements*) in **blue** (**not a rigorous derivation, just for insights!**)

$$\mathbf{g} = (g_1, g_2, \dots, g_K) \quad \mathbf{n} = (n_1, n_2, \dots, n_K) \quad n_i \sim \mathcal{N}(0, \frac{n_0}{2})$$



Apply the MAP Rule for the Vector

- Consider **binary antipodal** $a_i \in \{-1, 1\}$
- Need to compare the a posteriori probability

$$P(1|\mathbf{y}) = \frac{P(1)f(\mathbf{y}|1)}{f(\mathbf{y})} \begin{matrix} \hat{a} = 1 \\ > \\ \hat{a} = -1 \end{matrix} P(-1|\mathbf{y}) = \frac{P(-1)f(\mathbf{y}|-1)}{f(\mathbf{y})}$$

- With equal probability of -1 and 1, MAP is equivalent to ML

$$f(\mathbf{y}|1) \begin{matrix} \hat{a} = 1 \\ > \\ \hat{a} = -1 \end{matrix} f(\mathbf{y}|-1)$$



ML Rule

■ The conditional distributions:

$$f(\mathbf{y}|1) = \frac{1}{(\pi n_0)^{k/2}} \exp \left(- \sum_{i=1}^k \frac{(y_i - g_i)^2}{n_0} \right) = \frac{1}{(\pi n_0)^{k/2}} \exp \left(- \frac{\|\mathbf{y} - \mathbf{g}\|^2}{n_0} \right)$$

$$f(\mathbf{y}|-1) = \frac{1}{(\pi n_0)^{k/2}} \exp \left(- \frac{\|\mathbf{y} + \mathbf{g}\|^2}{n_0} \right)$$

■ The ML rule is equivalent to:

$$\frac{f(\mathbf{y}|1)}{f(\mathbf{y}|-1)} = \exp \left(\frac{-\|\mathbf{y} - \mathbf{g}\|^2 + \|\mathbf{y} + \mathbf{g}\|^2}{n_0} \right) = \exp \left(\frac{4\mathbf{y}\mathbf{g}^T}{n_0} \right) \begin{matrix} \hat{a} = 1 \\ > \\ < \\ \hat{a} = -1 \end{matrix} 1$$



Matched Filter is actually **Optimal** for SEP!

$$\frac{f(\mathbf{y}|1)}{f(\mathbf{y}|-1)} = \exp\left(\frac{-\|\mathbf{y} - \mathbf{g}\|^2 + \|\mathbf{y} + \mathbf{g}\|^2}{n_0}\right) = \exp\left(\frac{4\mathbf{y}\mathbf{g}^T}{n_0}\right)$$

$\hat{a} = 1$
 > 1
 < 1
 $\hat{a} = -1$

- The inner product of the two waveform vectors, when ***k* approaches infinity**, is actually equivalent to

$$\mathbf{y}\mathbf{g}^T \rightarrow \int_{-\infty}^{\infty} [a_i g(t) + n(t)] g(t) dt$$

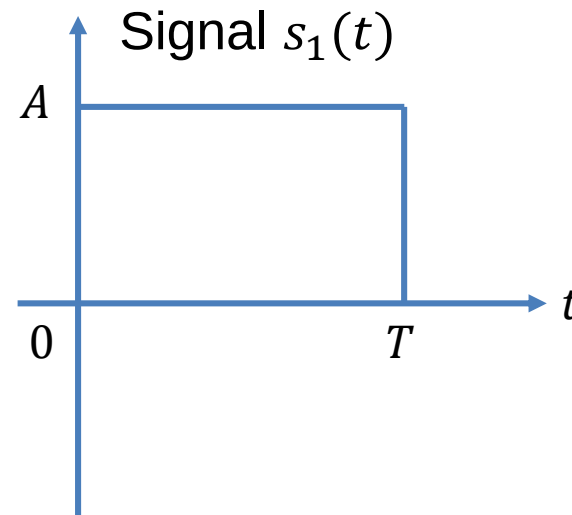
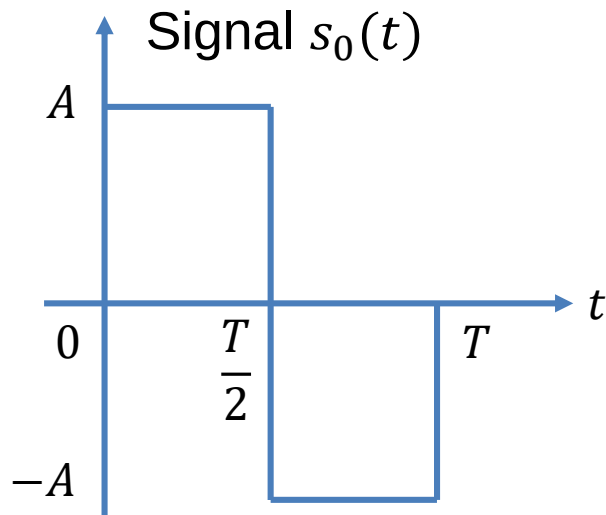
Correlator
receiver:
Matched Filter
output
sampled at T_s !

M.F. output is *sufficient* to apply the ML detection rule!
The receiver architecture we taught is actually optimal for SEP!



An Example of Baseband Transmission

- For a baseband transmission system with AWGN channel, transmit $s_0(t)$ if the bit is 0, transmit $s_1(t)$ if the bit is 1.



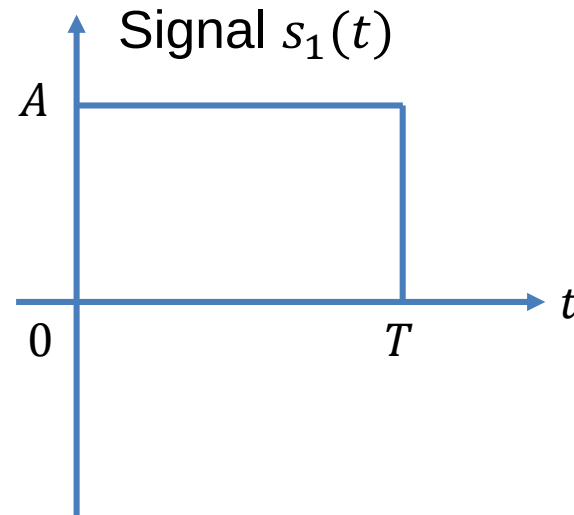
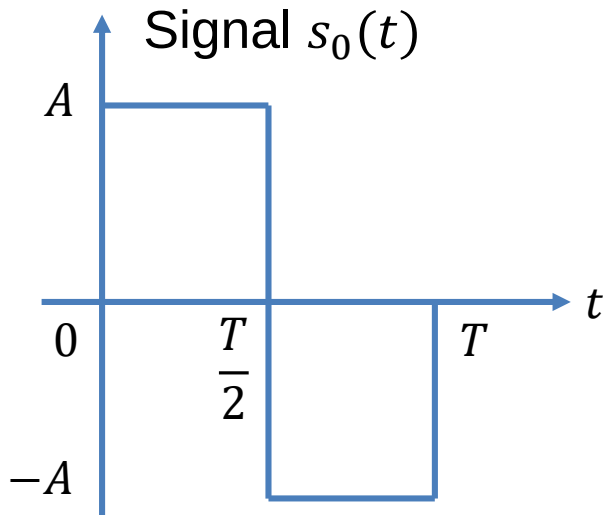
- The received signal is $r(t) = s_i(t) + z(t)$
 - ▶ $z(t)$ is Gaussian noise with auto-correlation

$$\Phi_{nn}(\tau) = E(n(t)n(t + \tau)) = \frac{1}{2}n_0\delta(\tau)$$



An Example of Baseband Transmission

- (a) Compute the energy of the impulse signal $s_0(t)$ and $s_1(t)$



$$E_0 = \int_0^T s_0^2(\tau) d\tau = A^2 T$$

$$E_1 = \int_0^T s_1^2(\tau) d\tau = A^2 T$$



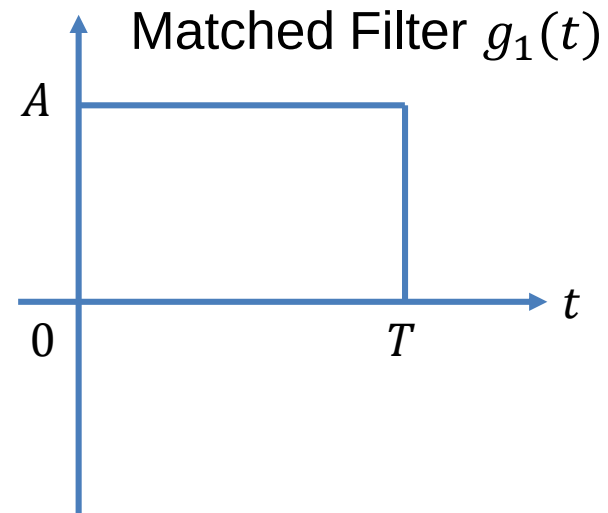
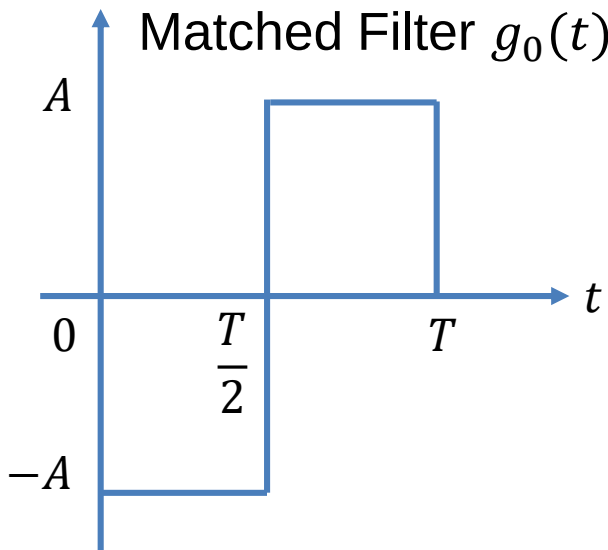
An Example of Baseband Transmission

- (b) The receiver is implemented by 2 matched filters, corresponding to $s_0(t)$ and $s_1(t)$ respectively.

Draw the impulse responses to the two matched filters.

$$g_0(t) = s_0(T - t)$$

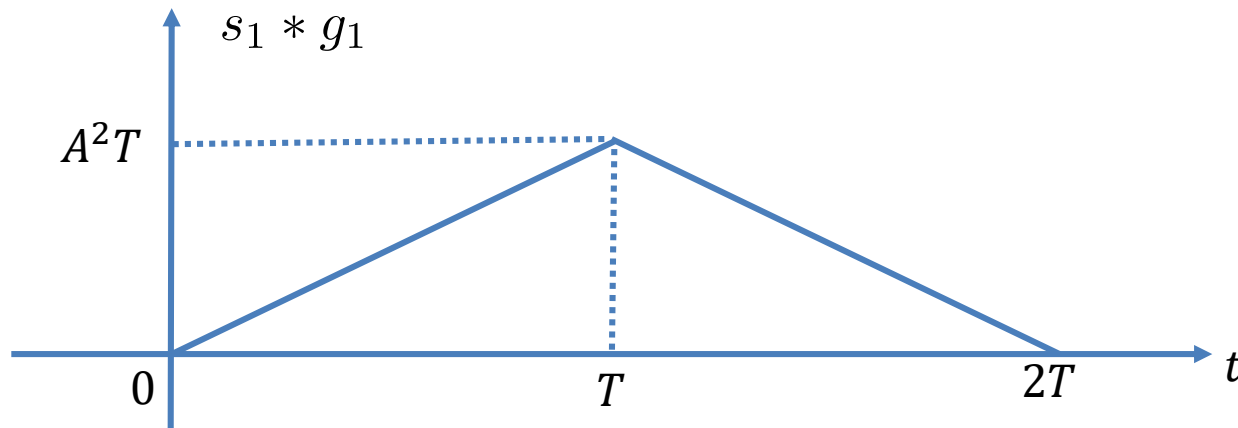
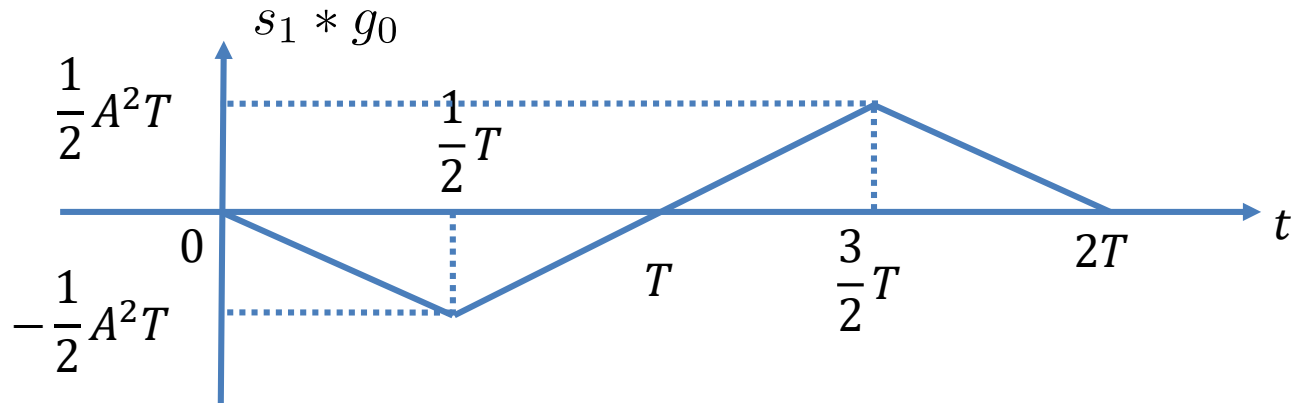
$$g_1(t) = s_1(T - t)$$





An Example of Baseband Transmission

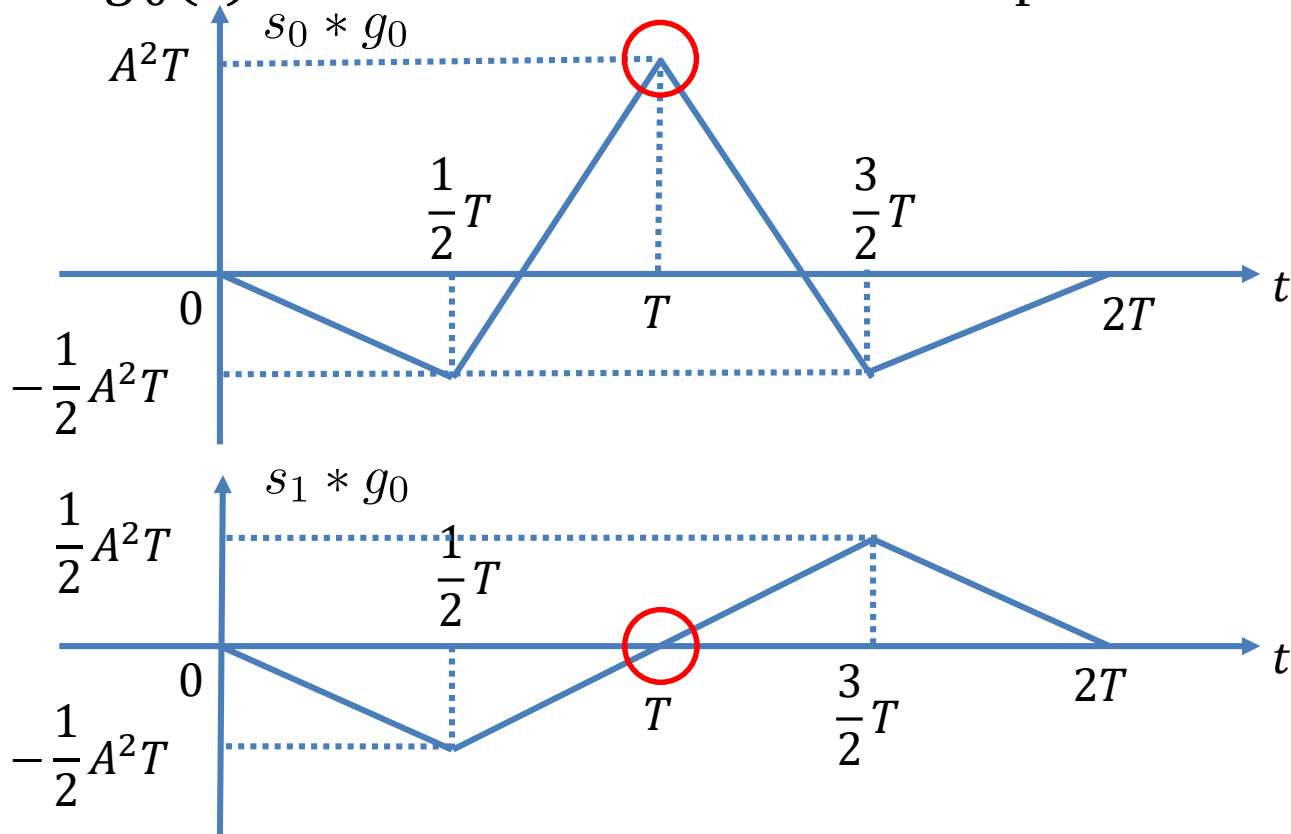
- (c) When transmitting signal $s_1(t)$, Draw the outputs for the two matched filters.





An Example of Baseband Transmission

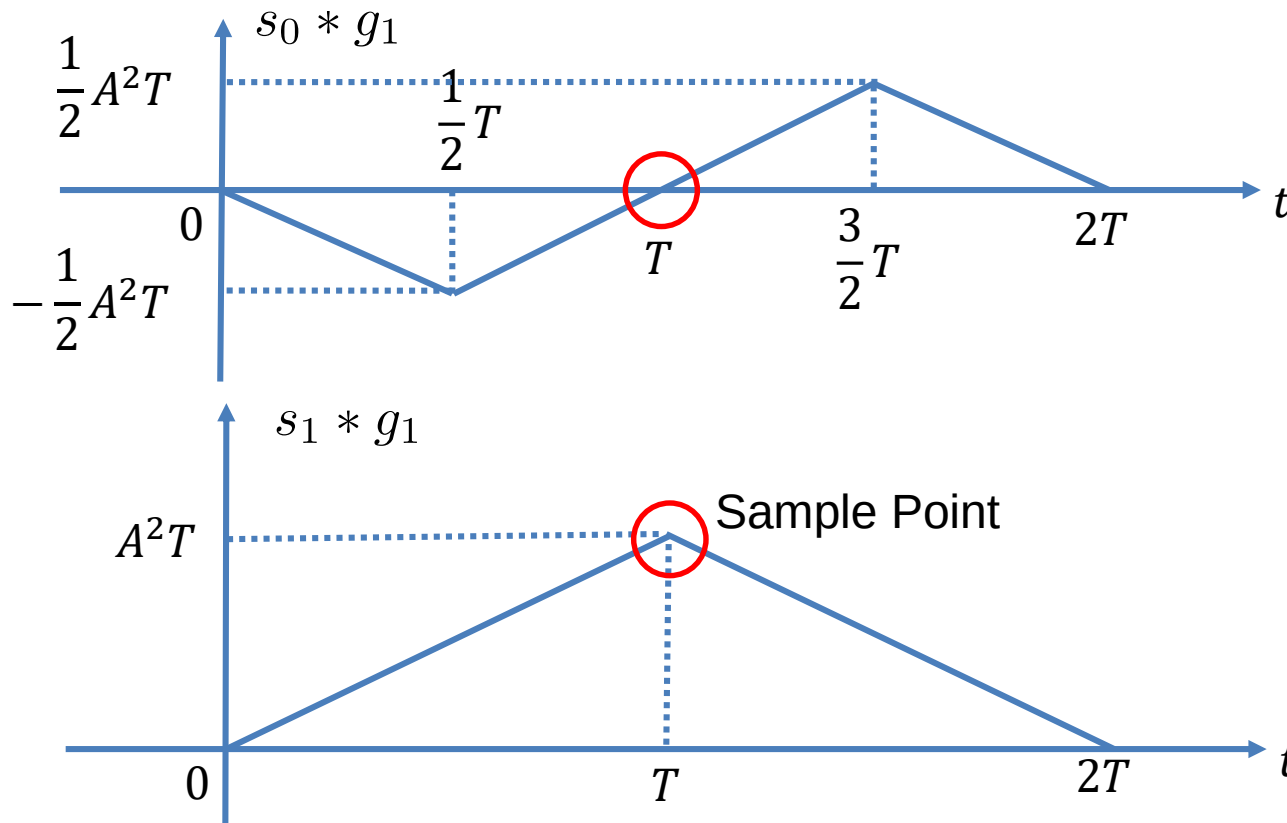
- (d) Describe the detection methods. Express the corresponding BER (Bit Error Rate).
- If we use $g_0(t)$ as the matched filter and sample at T :





An Example of Baseband Transmission

- (d) Describe the detection methods. Express the corresponding BER (Bit Error Rate).
- Or we use $g_1(t)$ as the matched filter and sample at T :





An Example of Baseband Transmission

- (d) Describe the detection methods. Express the BER (Bit Error Rate).
- The sampled outputs of one matched filter are $0, A^2T$, we can use the ML rule:

$$P_b = P_s = Q\left(\frac{A^2T / 2}{\sigma}\right)$$

- We know that the energy power at the sampling point after the filter is:

$$\mathbb{E}[n_o^2(T_s)] = R_{N_o}(0) = \int_{-\infty}^{\infty} S_{N_o}(f) e^{j2\pi f \cdot 0} df = \int_{-\infty}^{\infty} \frac{n_0}{2} |H(f)|^2 df$$

- Therefore:

$$P_b = P_s = Q\left(\frac{A^2T / 2}{\sqrt{\frac{n_0}{2}} A^2T}\right) = Q\left(\sqrt{\frac{A^2T}{2n_0}}\right)$$

Can we improve it?



Summary

- Optimal Detection ([Haykin: 4.3](#); Proakis: 5.1.3)
 - ▶ MAP + equal prob. Symbol $\rightarrow$ ML rule
 - ▶ ML rule + additive Gaussian Noise $\rightarrow$ Min. distance rule

- Error Probability Analysis ([Haykin: 4.3](#); Proakis: 5.2.1)
 - ▶ The **Q function** is a key element!

- Revisit the Baseband Receiver ([Haykin: 5.1-5.6](#),
**optional if you have time*)
 - ▶ MF + ML detector is optimal in terms of SEP